

► Introduction

In August 2023, the ILO published the first global employment estimates of potential occupational exposure to generative artificial intelligence (GenAI) (Gmyrek et al., 2023). This research was prompted by concerns – that have featured prominently in the media – of the potential for the technology to replace large shares of knowledge work.

Since the release of the 2023 estimates, there have been further technological advances in GenAI, including the introduction of Large Language Models (LLMs) with expanded capabilities in image recognition and audio and video production. Moreover, many new GenAI applications have been developed as a result of the possibility of linking GenAI models with Application Programming Interface (API) tools. Indeed, GenAI has been likened to a general-purpose technology, meaning that it may continue generating new applications and innovations across various sectors, driving long-term economic and societal transformations.

Despite this rapidly developing landscape, it is important for policy makers to develop deeper understanding of how technological advances might impact existing jobs. To address this need, the Research department of the ILO has partnered with NASK, the National Research Institute of the Ministry of Digital Affairs in Poland, to develop a more precise assessment of potential effects of GenAI on employment across countries.

This study builds on the method developed by the ILO (Gmyrek, Berg and Bescond, 2023 - GBB hereafter), which used task descriptions from the 4-digit International Standard Classification of Occupations (ISCO-08) and the GPT-4 model to estimate task-level and occupational scores of exposure to GenAI technology and then derives global employment estimates based on official ILO labour force data. But the study includes several key refinements.

The first step in our analysis is based on Poland's national 6-digit classification of occupations that includes nearly 30,000 tasks. This expands tenfold the number of tasks given in the ISCO-08 structure and enables a fine-grained assessment of tasks' automation potential, based on a more up-to-date set of occupations and tasks than those captured in the ISCO-08 system. As Poland's 6-digit classification is aligned with the 4-digit ISCO-08, the occupational scores can be compiled at the 4-digit level allowing a straightforward comparison and updating of the 2023 estimates of the ILO (GBB). In addition, the study improves the methodological framework of GBB, by drawing on the combined strengths of human and AI abilities, with an additional layer of expert verification.

Specifically, we surveyed 1,640 people currently employed in Poland in each ISCO-08 1-digit group to rank the automation potential of a representative sample of tasks that pertain to their occupational group. As the second step, we asked a mix of national and international experts to conduct a detailed review of a large sample of tasks previously assessed in the survey to validate or adjust the level of automation scores through an iterative process. Since we maintain the hierarchical link to ISCO-08 and focus on technological feasibility rather than specific country context, by combining opinions of local and international experts, we create a system that enables more precise predictions of the potential of automation of individual occupational tasks with GenAI technology, including at the international level. To achieve that, we input the human scores into an AI model, to generate scores reflecting human judgement for nearly 30,000 individual tasks pertaining to 2,500 occupations at 6-digit level in Poland. We then use the same AI predictor to re-generate scores for the tasks in ISCO-08, previously provided by GBB (2023), thereby adjusting the 2023 index of the ILO, which now benefits from both the input of people

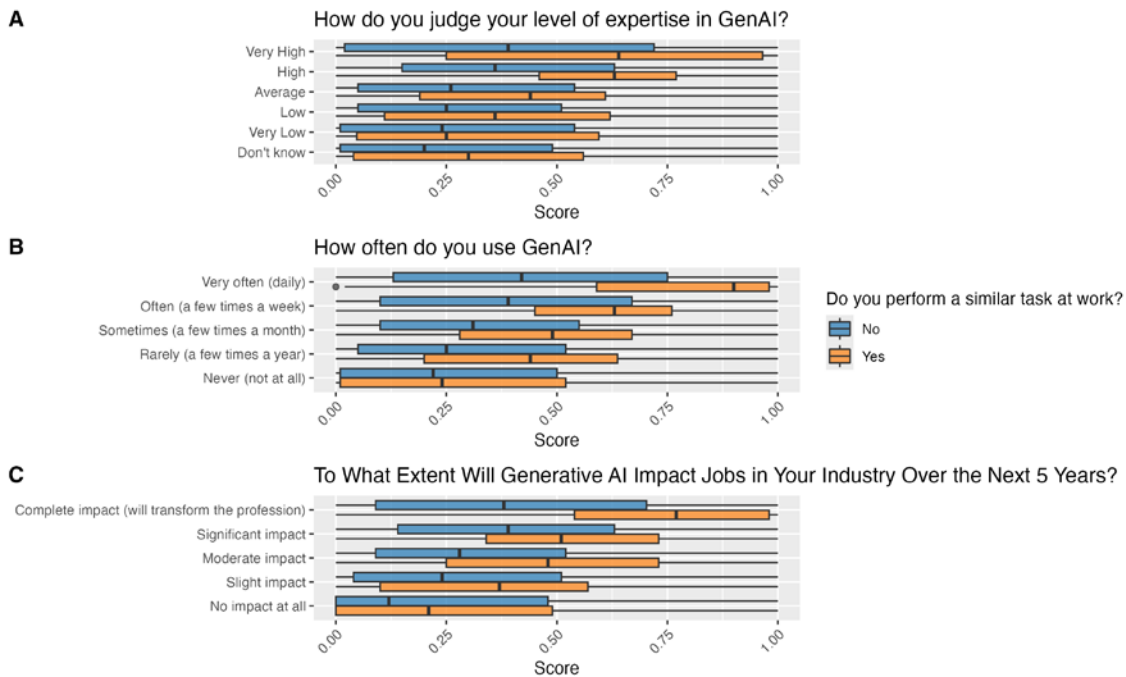
employed in each ISCO-08 1-digit group and from expert opinions. As the final step, we provide updated global, regional and income-based estimates of employment that has the potential to be transformed by Gen AI technology.

Our study specifically targets GenAI tools, isolating the effects of this recent family of AI from broader AI applications, such as the use of Machine Learning (ML) algorithms or image recognition in industrial production. There are several considerations that justify such selective focus. First, we respond to the growing need for a research tool that could provide a better picture of GenAI's potential impact on national labour markets, given the growing societal angst concerning this technology and the recent reports about increasing levels of adoption of GenAI at work (Bick et al., 2024; Faverio and Tyson, 2022; Grampp et al., 2023; Maison & Partners and ThinkTank, 2024; Pew Research Center, 2023; Rutgers, 2024). Second, while several indicators of occupational exposure to broader AI technologies are available to researchers (Nurski and Vansteenkiste, 2024), very few tools allow for a more detailed focus on GenAI (Nurski and Ruer, 2024). A new, more precise exposure index enables a closer alignment of the academic work on digital economy and labour with the significant interest in the GenAI technology in the media and in the public debate.¹ Third, the methodological blueprint provided by our research can be quite flexibly expanded to other types of AI in the future or focused more narrowly on specific subsets of digital technologies and sectors of their application.

This paper is the first in a series of upcoming publications based on ILO and NASK cooperation that strive to improve the methodology for assessing employment effects of AI, but also to document in detail the methods used, including the survey, expert verification process and the construction of the AI model used for predictions. Subsequent papers will present an assessment of the potential impact of GenAI on the Polish labour market, based on more detailed occupation and task-level data and using a mix of quantitative and qualitative methods. All publications will include the methodological blueprints and technical details of both the quantitative and qualitative components, including survey questionnaires and interview guidelines. By making these tools openly accessible, we hope to contribute to the development of an improved method for assessing potential impact of GenAI on jobs and to stimulate more detailed national-level studies, including in lower- and middle-income countries.

¹ For example, Felten et al. (2023) provided an updated version of their original exposure index from 2021, adjusting for the abilities of GenAI tools. As shown in their paper, the adjustment does not result in a major shift in occupations exposure and still makes it impossible to assess the abilities of this new technology in a selective manner. See Gmyrek et al. (2024) for a more detailed comparison of these scores.

► **Figure 10. Scoring as a function of familiarity with GenAI and the scored task – distribution of individual scores**



One way to address this variation in scoring would be to apply corrective weights to responses, giving greater influence to respondent profiles likely to have more relevant expertise. For example, one could assign more weight to respondents reporting greater familiarity with GenAI or frequent usage or prioritize scores from individuals who perform similar tasks in their daily work. However, in our sample, this would introduce significant limitations. For 91.5% of the tasks, respondents did not report performing a similar task in their work – even when the task belonged to the same 1-digit occupational group as their current job. This is not surprising. For instance, within the “Professionals” group alone, the representative sample consisted of 368 tasks drawn from a pool of 8,715 tasks (Table 2), making exact task matching unlikely.

Moreover, adjusting for beliefs or usage would require strong theoretical assumptions about what constitutes “credible” scoring. For example, a respondent who believes their profession will be completely transformed by GenAI in the next five years may be highly informed, or may be expressing anxiety, speculation, or dystopian fears. Without richer data to distinguish between these interpretations, applying corrective weights risks introducing new biases.

Given the relatively small size of these specific subgroups in our dataset, we refrain from implementing such adjustments in the current analysis to avoid overfitting or compromising credibility. Instead, we flag this as an important area for future survey design, particularly in capturing respondent expertise and task familiarity more systematically.

For the present study, we treat the survey as a starting point, designed to capture a broad spectrum of perspectives reflecting varying degrees of familiarity with both tasks and GenAI. In many cases, a respondent’s understanding of the task itself may be more influential than their understanding of the technology, especially for concrete physical tasks (e.g. trimming flowers, assembling mechanical parts). Accordingly, we rely on mean scores across all respondents to reflect the collective judgment of a diverse sample grounded in workplace experience. To enhance

robustness, we supplement this with expert evaluation of a subset of tasks (section 5), ensuring that final scores reflect both practical insights and a realistic appraisal of current GenAI capabilities.

resource-intensive and limited by our relatively small dataset.¹⁵ In contrast, by using a sufficiently large set of example tasks within the prompt, we can provide the model with explicit “demonstrations” of scoring of similar tasks in our main dataset. To ensure a balanced approach, we provide each prompt with a sample of tasks with the highest 50 and lowest 50 scores, alongside the scores of 30 tasks with greatest semantic similarity to the tasks that the model is requested to score. To create the semantic context, we construct a rapid-search vector database with all 2,861 adjusted scores. The database is queried at the start of each API call and extracts tasks with semantic similarity to the task that is being scored. Once the 30 most similar tasks have been retrieved, a set of complete instructions is passed to the model's API, as per the general prompt strategy illustrated in Box 3.

We first validate the predictive model using a sample of tasks from the survey that were adjusted as per sections 5-7, but not included in the semantic context provided to the model.¹⁶ Once the prompt adjustments deliver satisfactory performance, we use the same system to predict automation scores for 3,265 tasks in ISCO-08 (as in Gmyrek et al., 2023).¹⁷ This approach bridges expert-validated human assessments with the scalability of LLM-based predictions, enabling consistent and extensible scoring for tasks across classification systems.

For a further refinement of our scoring system, we filter 364 tasks where the scoring in 2025 changed by more than 25% of the absolute value of 2023 score. We review these tasks manually, using an expert consensus-building method. First, we conduct a mini survey among co-authors, in which each respondent is asked to arbitrate between the diverging scores. For 64 ISCO-08 tasks, where we still observe a divergence of expert opinions, we conduct a detailed group discussion, in order to reach consensus and agree on a justification.

Following the final manual verification process, we establish a second knowledge repository, also stored in a vector database. The main LLM prompt is then updated to incorporate this additional reference by conducting two sequential semantic context retrievals for each prediction, further improving the model's precision in predicting scores for new tasks. Using this approach, we generate predictions for all 29,753 tasks in the Polish classification system (Table 1).

To compare these 6-digit predictions with 4-digit ISCO-08 classifications, we calculate (i) the mean of all task scores aggregated at the 4-digit level and (ii) occupational means at the 6-digit level, subsequently deriving the 4-digit mean from these 6-digit components. In both cases, the correlation between predictions based on ISCO-08 tasks and Polish 6-digit tasks is 0.92, demonstrating exceptional stability of predictions. In other words, while one might question the exact scores assigned to tasks in our knowledge repository – a challenge not unique to our study – once the repository and prompting techniques are in place, the model consistently produces stable predictions based on reference tasks and their scores, and irrespective of the occupational classification system. This approach offers promising possibilities for replication across other occupational classifications, which we intend to explore further. For the remainder of this paper, we focus primarily on revising ISCO-08 scores, leaving a detailed analysis of Poland's 6-digit occupations for a separate study (Gmyrek et al., forthcoming, 2025).

¹⁵ See OpenAI fine-tuning guidelines and developer forum <https://platform.openai.com/docs/guides/fine-tuning>.

¹⁶ These tasks were split into an 80-20 proportion, with 80% used for training and 20% reserved for validation. Performance was measured using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE), with the adjusted scores showing better agreement (MAE = 0.0494, RMSE = 0.0724) compared to the predicted scores (MAE = 0.0684, RMSE = 0.0963). Correlation analysis further supported this, with adjusted scores exhibiting higher Pearson (0.895) and Spearman (0.884) correlations than predicted scores (Pearson: 0.796, Spearman: 0.776). This is understandable, as the adjusted scores were based on two points of reference (survey score and expert score), making them more stable and easier to predict. In contrast, for the predicted scores, the test involves a repeated prediction process, which amplifies the original error.

¹⁷ There are 3,123 distinct tasks in ISCO-08 and 3,265 tasks in total, as some occupations share the same task.